Polar Curves and Derivatives

ANSWER KEY

Section 1: Slopes of Tangent Lines to Polar Curves

1 Which formula gives the slope $\frac{dy}{dx}$ of the tangent line to the polar curve $r = f(\theta)$?

a. $\frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$ ✓

b. $\frac{dr}{d\theta}$

c. $\frac{r\sin\theta}{r\cos\theta}$

d. $\frac{\frac{dr}{d\theta}\cos\theta}{\frac{dr}{d\theta}\sin\theta}$

From $x = r\cos\theta$ and $y = r\sin\theta$, applying product rule: $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$.

2 Find the slope of the tangent line to the circle $r = 2\cos\theta$ at $\theta = \frac{\pi}{4}$.

a. -1 ✓

b. 1

c. 0

d. Undefined

At $\theta = \pi/4$: $r = 2(\frac{\sqrt{2}}{2}) = \sqrt{2}$. $r' = -2\sin\theta = -\sqrt{2}$. Numerator $= -\sqrt{2}(\frac{\sqrt{2}}{2}) + \sqrt{2}(\frac{\sqrt{2}}{2}) = -1 + 1 = 0$. Denominator $= -\sqrt{2}(\frac{\sqrt{2}}{2}) - \sqrt{2}(\frac{\sqrt{2}}{2}) = -1 - 1 = -2$. Slope $= \frac{0}{-2} = 0$ (or at $\theta = \pi/4$: horizontal tangent).

3 What is the slope of the tangent line to the cardioid $r = 1 + \sin\theta$ at $\theta = 0$?

a. 1 ✓

b. -1

c. 0

d. Undefined

At $\theta = 0$: $r = 1$. $r' = \cos(0) = 1$. Numerator $= 1(0) + 1(1) = 1$. Denominator $= 1(1) - 1(0) = 1$. Slope $= \frac{1}{1} = 1$.

Section 2: Horizontal and Vertical Tangents on Polar Curves

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4 Horizontal tangent lines to a polar curve occur where which condition is met?

a. $\frac{dy}{d\theta} = 0$ **and** $\frac{dx}{d\theta} \neq 0$ ✓

b. $\frac{dr}{d\theta} = 0$

c. $\frac{dx}{d\theta} = 0$

d. $r = 0$

A horizontal tangent requires the numerator of $\frac{dy}{dx}$ to be zero: $\frac{dy}{d\theta} = 0$ (provided $\frac{dx}{d\theta} \neq 0$).

5 Vertical tangent lines occur where which condition is met?

a. $\frac{dx}{d\theta} = 0$ **and** $\frac{dy}{d\theta} \neq 0$ ✓

b. $\frac{dr}{d\theta} = 0$

c. $\frac{dy}{d\theta} = 0$

d. $r = 1$

A vertical tangent requires the denominator of $\frac{dy}{dx}$ to be zero: $\frac{dx}{d\theta} = 0$ (provided $\frac{dy}{d\theta} \neq 0$).

6 At the pole ($r = 0$), what is the slope of the tangent line to the curve $r = f(\theta)$ when $f(\alpha) = 0$?

a. $\tan\alpha$ ✓

b. $\cot\alpha$

c. 0

d. Undefined

When $r = 0$, $\frac{dy}{dx} = \frac{r'\sin\alpha + 0}{r'\cos\alpha - 0} = \frac{\sin\alpha}{\cos\alpha} = \tan\alpha$.

Section 3: Rate of Change of r and Distance from Origin

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7 For the polar curve $r = 3 - 2\cos\theta$, what is $\frac{dr}{d\theta}$ at $\theta = \frac{\pi}{2}$?

a. 2 ✓

b. -2

c. 0

d. 3

$\frac{dr}{d\theta} = -2(-\sin\theta) = 2\sin\theta$. At $\theta = \frac{\pi}{2}$: $2\sin(\frac{\pi}{2}) = 2(1) = 2$.

8 If $\frac{dr}{d\theta} > 0$ for a polar curve, what does this indicate about the distance from the pole?

a. **The distance from the origin is increasing as θ increases** ✓

b. The distance is decreasing

c. The curve is concave up

d. The curve is a line

Since r measures directed distance from the origin, $\frac{dr}{d\theta} > 0$ (for $r > 0$) means the particle is moving away from the pole.

9 At what angle $\theta \in [0, 2\pi)$ is the distance from the origin maximized for $r = 2 + 3\cos\theta$?

a. $\theta = 0$ ✓

b. $\theta = \pi$

c. $\theta = \frac{\pi}{2}$

d. $\theta = \frac{3\pi}{2}$

Maximum value of $\cos\theta$ is 1, which occurs at $\theta = 0$, giving maximum distance $r = 2 + 3(1) = 5$.

Section 4: Arc Length of Polar Curves

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10 Which integral represents the arc length of the polar curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$?

a. $\int_{\alpha}^{\beta} \sqrt{r^2 + (\frac{dr}{d\theta})^2}\, d\theta$ ✓

b. $\int_{\alpha}^{\beta} \sqrt{1 + (\frac{dr}{d\theta})^2}\, d\theta$

c. $\frac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta$

d. $\int_{\alpha}^{\beta} r d\theta$

By transforming $ds = \sqrt{(dx)^2 + (dy)^2}$, polar arc length is $L = \int_{\alpha}^{\beta} \sqrt{r^2 + (r')^2}\, d\theta$.

11 Find the total perimeter of the circle $r = 4\cos\theta$ for $\theta \in [0, \pi]$.

a. 4π ✓

b. 8π

c. 2π

d. 16π

$r' = -4\sin\theta$. $\sqrt{r^2 + (r')^2} = \sqrt{16\cos^2\theta + 16\sin^2\theta} = \sqrt{16} = 4$. Perimeter $= \int_0^{\pi} 4 d\theta = 4\pi$.

12 Set up the arc length of the cardioid $r = 1 + \cos\theta$ from $\theta = 0$ to $\theta = 2\pi$.

a. $\int_0^{2\pi} \sqrt{2 + 2\cos\theta}\, d\theta$ ✓

b. $\int_0^{2\pi} (1 + \cos\theta) d\theta$

c. $\int_0^{2\pi} \sqrt{1 + \cos^2\theta}\, d\theta$

d. $2\int_0^{\pi} (1 + \sin\theta) d\theta$

$r^2 + (r')^2 = (1 + \cos\theta)^2 + (-\sin\theta)^2 = 1 + 2\cos\theta + \cos^2\theta + \sin^2\theta = 2 + 2\cos\theta$.

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